

Two undergraduate projects: stratified sampling for redistricting plans

June 11, 2026

1 Context

These projects build on *Towards stratified sampling for redistricting plans* (Wang, Herschlag, Yim, Mattingly, Gilbert), which constructs a stratification of the space of redistricting plans: districts pooled from an ensemble are clustered into representative “letters” with HCCULTRAFIT [2], plans are encoded as unordered tuples of letters (“words”), and a partition of unity over a representative word set yields stratum weights z_s and a flux matrix F . The paper stops short of two things: actually running stratified (umbrella) sampling on these strata, and connecting the strata to the election outcomes that ensembles are used to study in litigation [3]. Each project takes one of these gaps.

Both projects work at two scales.

Small scale (exact): the Duplin–Onslow system. A 25-unit dual graph (24 Onslow County, NC precincts plus Duplin County as a single unit) from the Duke Quantifying Gerrymandering project, with real population, Black voting-age population, and precinct returns for 22 races (2008–2016) attached, together with an exhaustive enumeration of *all* 17,653 population-balanced contiguous 3-district plans. Because the plan space is fully enumerated, every pipeline quantity — letters, words, POU, z_s , F , and every election observable — is exact. There is no sampling error anywhere: every claim is checkable against ground truth. The full pipeline already runs on this system in the `gmstrat` repository (see Section 5).

Full scale (sampled): Connecticut and North Carolina. The real-state pipelines from the paper ($|V| = 462$, $I = 5$ and $|V| = 2671$, $I = 14$), sampled with Metropolized Cycle Walk [1], plus the $m \times n$ grid-graph ladder in `gmstrat` (`data/graph/`, 6×6 through 20×20) for controlled scaling experiments between the two regimes.

Established starting points (computed June 2026, exact)

On Duplin–Onslow, with the population-weighted symmetric-difference metric, $K = 12$ letters, beam width $L = 10$ (49 strata):

- The 14,345 distinct districts fill district space densely: within-cluster dispersion only halves by $K = 80$, with the interpretable structure exhausted by $K \approx 10$ –12. Letters at $K = 12$ are geographically legible motifs (a Duplin-anchored type, Jacksonville-centered, coastal-south, northern-tier).

- The strata communicate well: exact flux matrix with spectral gap $1 - |\lambda_2| \approx 0.48$; largest exact stratum weight $z_s \approx 0.074$.
- **Alignment with election outcomes is weak.** The between-stratum share of variance (an alignment R^2) for seat counts is 1–12% under the soft POU; with hard nearest-word assignment it reaches only ≈ 0.26 – 0.50 at $K = 80$, where the 1,038 strata hold ≈ 17 plans each.
- **Rare events are localized.** Six 2008 races have genuine seat variation; four races are rare events (a Democratic seat exists in 0.1–1% of plans). At $K = 12$ the top-3 strata contain 91–100% of the rare-event plans, even though per-stratum enrichment is modest. All 2010–2016 races are flat (0 Democratic seats in every plan) — no line-drawing choice can change them.

These numbers frame both projects: there is a lot of alignment R^2 “on the table” for Project 1, and Project 2 has exact rare-event probabilities to benchmark against.

2 Project 1: What does the stratification see?

Alignment of geometric strata with election outcomes

Question

The pipeline’s strata are built from geometry alone. Are they informative about the observables ensembles are actually used for — seat counts, partisan measures [6, 7], majority-minority structure? If not, can election-aware district metrics produce better-aligned strata without destroying spatial interpretability?

Why the question matters

Stratified sampling reduces the variance of $\hat{\mathbb{E}}_\pi[f]$ only through the between-stratum component of $\text{Var}_\pi(f)$. If the strata are nearly independent of f , stratification cannot help, no matter how well the sampler is engineered. The alignment R^2 is therefore the design quantity for the whole program, and the Duplin–Onslow numbers above show it is small for the natural geometric metric. Whether that is intrinsic (seat outcomes depend on fine boundary details that any coarse geometric summary discards) or fixable (a vote-aware metric recovers it) is open, and either answer matters.

Stage 1 — exact decomposition on Duplin–Onslow

For each observable f (six competitive 2008 seat counts, four rare-event indicators, max district BVAP share), decompose

$$\text{Var}_\pi(f) = \underbrace{\sum_s z_s (\mathbb{E}_{\pi_s}[f] - \mathbb{E}_\pi[f])^2}_{\text{between}} + \underbrace{\sum_s z_s \text{Var}_{\pi_s}(f)}_{\text{within}}, \quad \frac{d\pi_s}{d\pi} = \frac{\phi_s}{z_s},$$

exactly, sweeping the POU temperature T , beam width L , and alphabet size K . Reproduce and extend the baseline table in `local/duplin/alignment_R2_K12_L10.csv`. Characterize the R^2 -vs- (number of strata) tradeoff curve and the rare-event concentration $\max_s P(\text{event} \mid s)$. This stage is mostly running and interrogating existing code; first plots within two weeks.

Stage 2 — election-aware letters

Augment the district distance with a vote-share term,

$$d_\lambda(u, v) = d_{\text{geom}}(u, v) + \lambda W |\text{share}(u) - \text{share}(v)|,$$

re-run HCCULTRAFIT and the word construction across λ (minutes of compute at this scale), and trace alignment-vs- λ for each observable. The tension to characterize: as λ grows, do letters remain spatially interpretable district types, or fragment into partisan bins? Quantify interpretability (e.g., spatial coherence of letter densities) so the tradeoff curve has two real axes.

Stage 3 — does it transfer? (full scale)

Repeat Stage 1 on the Connecticut ensemble from the paper (50K Cycle Walk samples, $K = 34$, statewide vote overlays from VEST), where the decomposition is no longer exact but the sample is large. Compare: is the weak-alignment finding a quirk of the lopsided Duplin–Onslow electorate (all 2010–2016 races are flat there), or does it persist in a competitive state? This stage turns the exact small-scale finding into a paper-ready claim.

Deliverables and skills

A reproducible alignment analysis (notebook + figures) for both scales; an alignment section that slots into the follow-up paper. Requires Python/pandas and basic probability (law of total variance); no MCMC internals. The pipeline is used as a black box.

3 Project 2: Does stratified sampling work? Implementing and benchmarking the umbrella sampler

Question

Implement the stratified sampler the paper sets up — tilted chains per stratum, EMUS recombination [4, 5] — and benchmark it where the right answer is known exactly.

Why Duplin–Onslow first

A sampler benchmark is only as good as its ground truth. Here the exact stratum weights z_s , the exact flux matrix, and the exact rare-event probabilities (e.g. $P(\text{EL08G_PR Dem seat}) \approx 0.002$) are already computed. Every component of the implementation can be unit-tested against truth before any scaling question arises. The $I = 3$ district matching is also trivial at this scale, so the sampler logic can be developed where the bookkeeping is easy.

Stage 1 — pin down the tilt

The draft writes $J_s = J - \phi_s$, which is at most an e -fold reweighting (since $\phi_s \in [0, 1]$) — too weak to confine a chain to a stratum. The standard umbrella choice is $J_s = J - \log \psi(x; s)$, i.e. $\pi_s \propto \psi(\cdot; s) \pi$, with unnormalized ψ avoiding the POU normalizer in the Metropolis–Hastings ratio. Resolving this (with the group) is the first milestone and a genuine contribution to the writeup.

Stage 2 — biased chains on Duplin–Onslow

Modify the Cycle Walk acceptance step to include $\psi(x'; s)/\psi(x; s)$. Evaluating ψ requires the plan-to-word distance — a min-cost bipartite matching [8] that can be warm-started, since each move changes few districts. Run one chain per stratum, estimate the overlap matrix $\hat{F}$, solve $z^\top F = z^\top$, and attach EMUS asymptotic error bars [4]. **Validation gates:** $\hat{z}_s \rightarrow z_s$ (exact), $\hat{F} \rightarrow F$ (exact), estimated observables $\rightarrow$ exact values, with error bars that cover at the nominal rate.

Stage 3 — the benchmark

Headline plot: MSE vs. total compute for the stratified estimator against a single long Cycle Walk chain, for (a) a smooth observable (max BVAP share, or a competitive 2008 seat count) and (b) a rare-event probability, with the exact answer drawn as a horizontal line. The rare events are where the win should be dramatic: the top-3 strata contain 91–100% of the rare-event plans, so chains tilted into those strata see the event constantly while the unstratified chain waits for a 10^{-2} – 10^{-3} fluctuation.

Stage 4 — climb the ladder (full scale)

Move up the grid-graph sequence (6×6 through 20×20 , pre-generated in `data/graph/`) to where enumeration is impossible and stratification has to earn its keep, tracking how the matching cost, overlap structure, and variance reduction scale with $|V|$ and I . Connecticut ($I = 5$) is the graduation exercise and the bridge back to the paper; the incremental-matching machinery developed here is what makes $I = 14$ (North Carolina) feasible later.

Deliverables and skills

A working, validated stratified sampler in the `gmstrat` repository; the MSE-vs-compute benchmark at both scales; a methods section for the follow-up paper. Requires stronger programming (Python plus some Julia for the Cycle Walk harness) and comfort with MCMC. Main schedule risk is ramping up on the Cycle Walk codebase — pair with Zijian early.

4 How the projects interlock

Project 1 tells Project 2 which observables stratification can plausibly help with: if geometric strata cannot see seat counts, Project 2 benchmarks against Project 1’s election-aware strata, and the comparison itself is the joint result. Both run on infrastructure that exists today; neither blocks the other. The natural merge point is a follow-up paper — stratified sampling for election-outcome estimation, validated exactly at small scale — with both students on it.

5 Infrastructure inventory

All in the `gmstrat` repository (branch `grid-experiments`):

- `duplin.py` — Duplin–Onslow converters, a fast HCCULTRAFIT implementation (validated to produce byte-identical linkage against `hccfit.py`; full 14,345-district run in ≈ 3 minutes), election utilities.
- `data/duplin-onslow/` — raw Duke data, gmstrat-format graph, all 17,653 plans in Cycle Walk JSONL format.
- `local/duplin/` — cached pipeline artifacts (distance matrix, linkage, per-plan seat counts), the elbow and stratification drivers (`run_elbow.py`, `run_strata.py`), figures, and the baseline alignment table.
- `data/graph/` — the grid ladder; `sampling/` — the Julia Cycle Walk harness; `demo.ipynb` — the canonical end-to-end pipeline walkthrough.
- The annotated reading list (with per-project reading orders) lives in the Overleaf project under `undergrad_reading/`.

References

- [1] D. R. DeFord, G. Herschlag, J. C. Mattingly, *A cycle walk for sampling measures on spanning forests for redistricting*, arXiv:2509.08629, 2025.
- [2] J.-H. Yim, A. C. Gilbert, *Fitting trees to ℓ_1 -hyperbolic distances*, arXiv:2409.01010, 2024.
- [3] G. Herschlag, H. S. Kang, J. Luo, C. V. Graves, S. Bangia, R. Ravier, J. C. Mattingly, *Quantifying gerrymandering in North Carolina*, Statistics and Public Policy 7(1):30–38, 2020.
- [4] E. H. Thiede, B. Van Koten, J. Weare, A. R. Dinner, *Eigenvector method for umbrella sampling enables error analysis*, J. Chem. Phys. 145, 084115, 2016.
- [5] A. R. Dinner, E. H. Thiede, B. Van Koten, J. Weare, *Stratification as a general variance reduction method for Markov chain Monte Carlo*, SIAM/ASA J. Uncertainty Quantification 8(3):1139–1188, 2020.
- [6] N. O. Stephanopoulos, E. M. McGhee, *Partisan gerrymandering and the efficiency gap*, University of Chicago Law Review 82(2):831–900, 2015.
- [7] J. N. Katz, G. King, E. Rosenblatt, *Theoretical foundations and empirical evaluations of partisan fairness in district-based democracies*, American Political Science Review 114(1):164–178, 2020.
- [8] R. Burkard, M. Dell’Amico, S. Martello, *Assignment Problems*, SIAM, 2009.